

HC-SR04 Ultrasonic Sensor



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Professors:

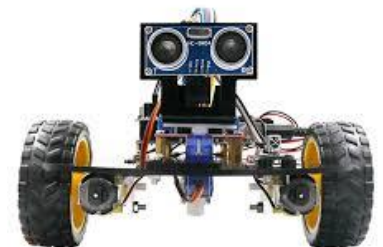
Safat Khan & Reda Fayek

5/10/25

Introduction

The HC-SR04 ultrasonic sensor measures distance using high-frequency sound waves and echo detection. It offers accurate, non-contact measurements from 2 cm to 400 cm, making it ideal for robotics, obstacle avoidance, and automation. With its simple 4-pin interface, it's easy to integrate into microcontroller-based systems for real-time distance sensing.

The HC-SR04 ultrasonic sensor is a compact and reliable module that allows it to be easily mounted on mobile robots for real-time collision detection and navigation.



Working Principle

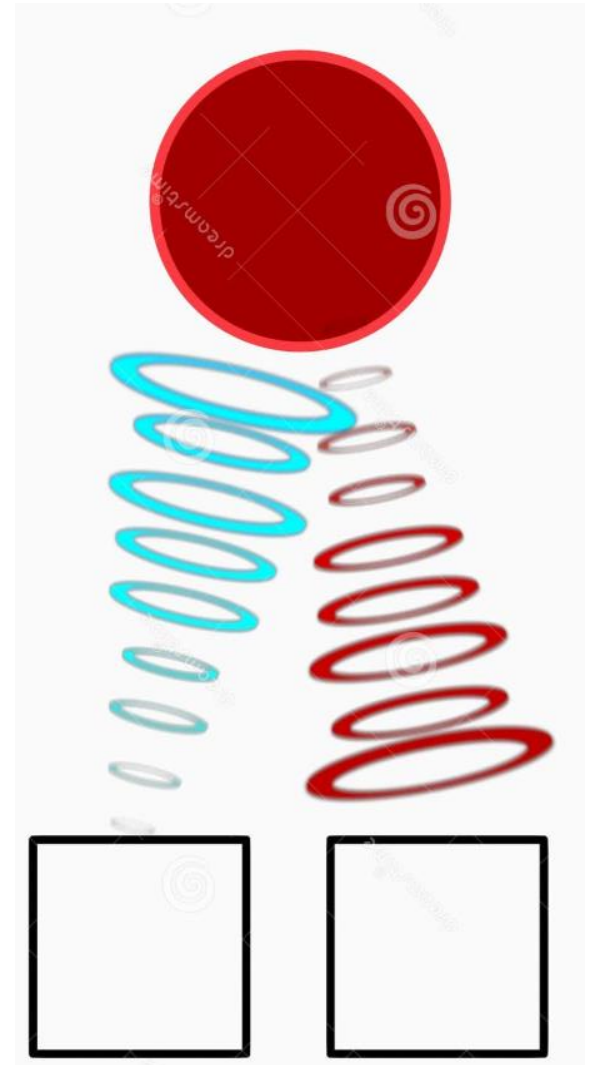
- Distance can be calculated as:

$$\text{Distance (cm)} = \frac{\text{Echo Time } (\mu\text{s})}{58}$$

or

$$\text{Distance (m)} = \frac{\text{Echo Time} \times 340}{2}$$

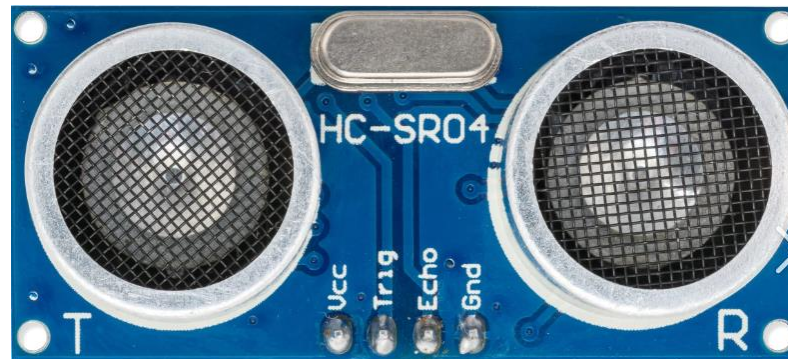
- Distance = $\frac{v \times \text{Echo Time}}{2}$
- where $v = 340 \text{ m/s}$ (speed of sound in air).
- Convert units:**
- $340 \text{ m/s} = 34,000 \text{ cm/s} = 0.034 \text{ cm}/\mu\text{s}$
- Distance (cm) = $\frac{0.034 \times \text{Echo Time } (\mu\text{s})}{2} = 0.017 \times \text{Echo Time } (\mu\text{s})$
- Simplify constant:**
- $\frac{1}{0.017} \approx 58.8 \Rightarrow \boxed{\text{Distance (cm)} = \frac{\text{Echo Time } (\mu\text{s})}{58}}$



Sensor Specifications

Parameter	Specification
Operating Voltage	5V DC
Operating Current	15 mA
Operating Frequency	40 kHz
Measurement Range	2 cm – 400 cm
Accuracy	±3 mm
Measuring Angle	15°
Trigger Input Signal	10 μs TTL pulse
Echo Output Signal	TTL-level signal proportional to distance
Dimensions	45 × 20 × 15 mm

Pinout and Functionality



VCC: Power
Input 5V

Trigger: Sends 10
 μ s pulse to initiate
measurement.

Echo – Returns
pulse.

GND: Ground

Test Setup: Diagram & Real Image

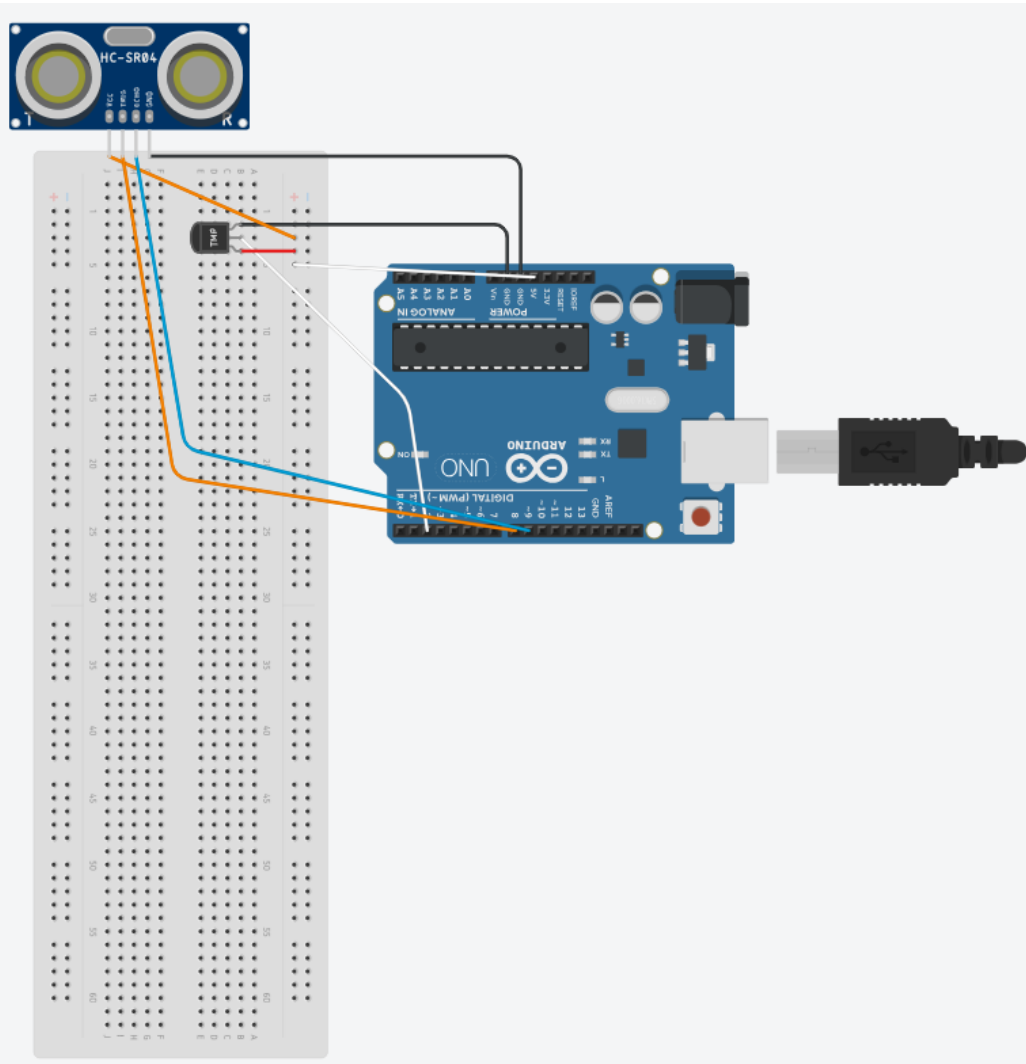
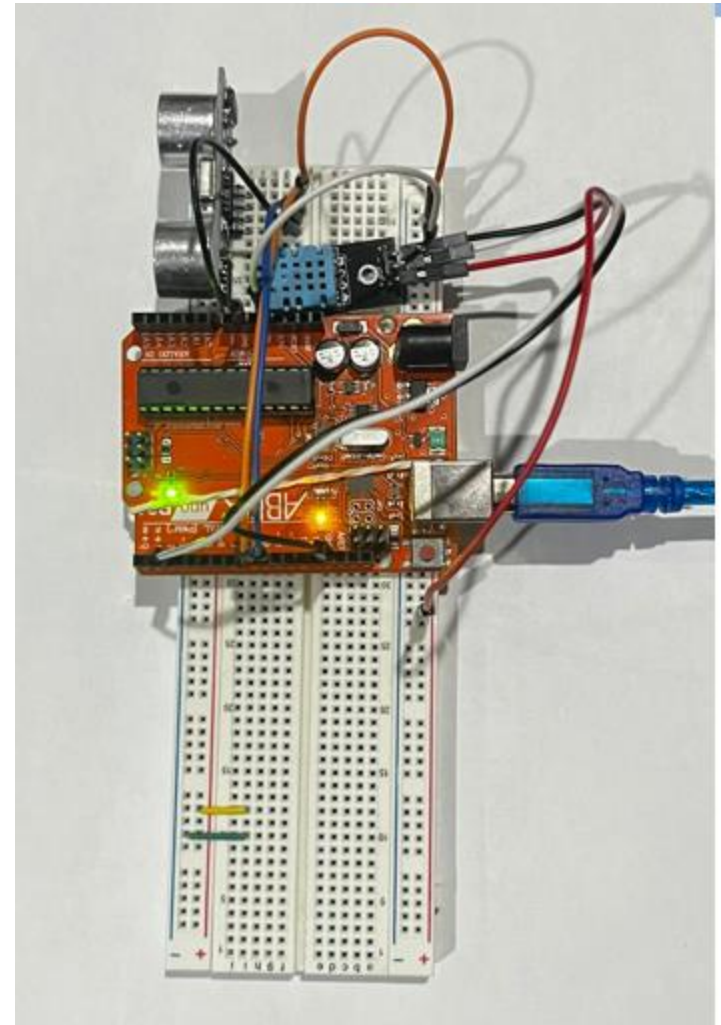
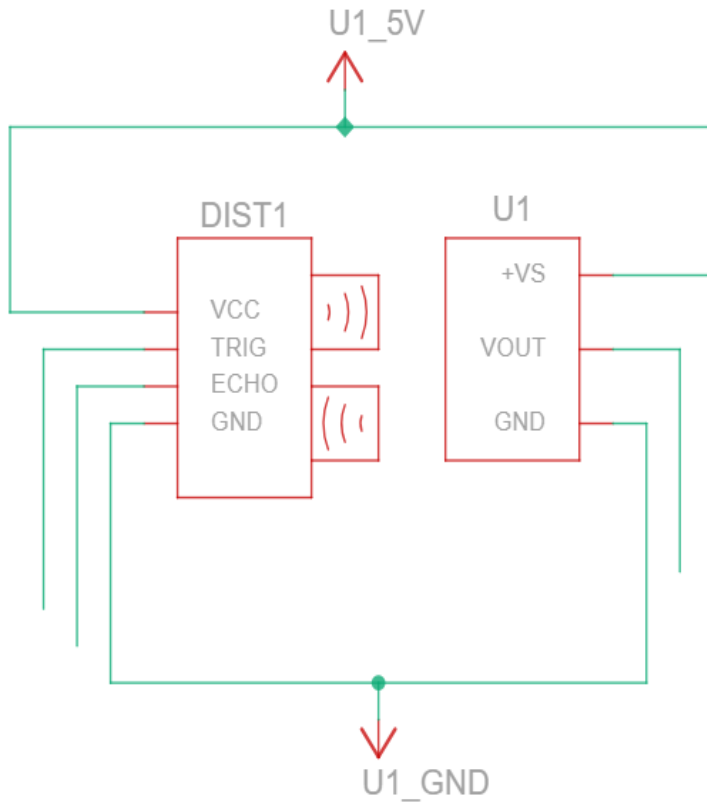


Diagram made using Tinkercad



Real image of test setup

Test Setup: Schematic & Wiring

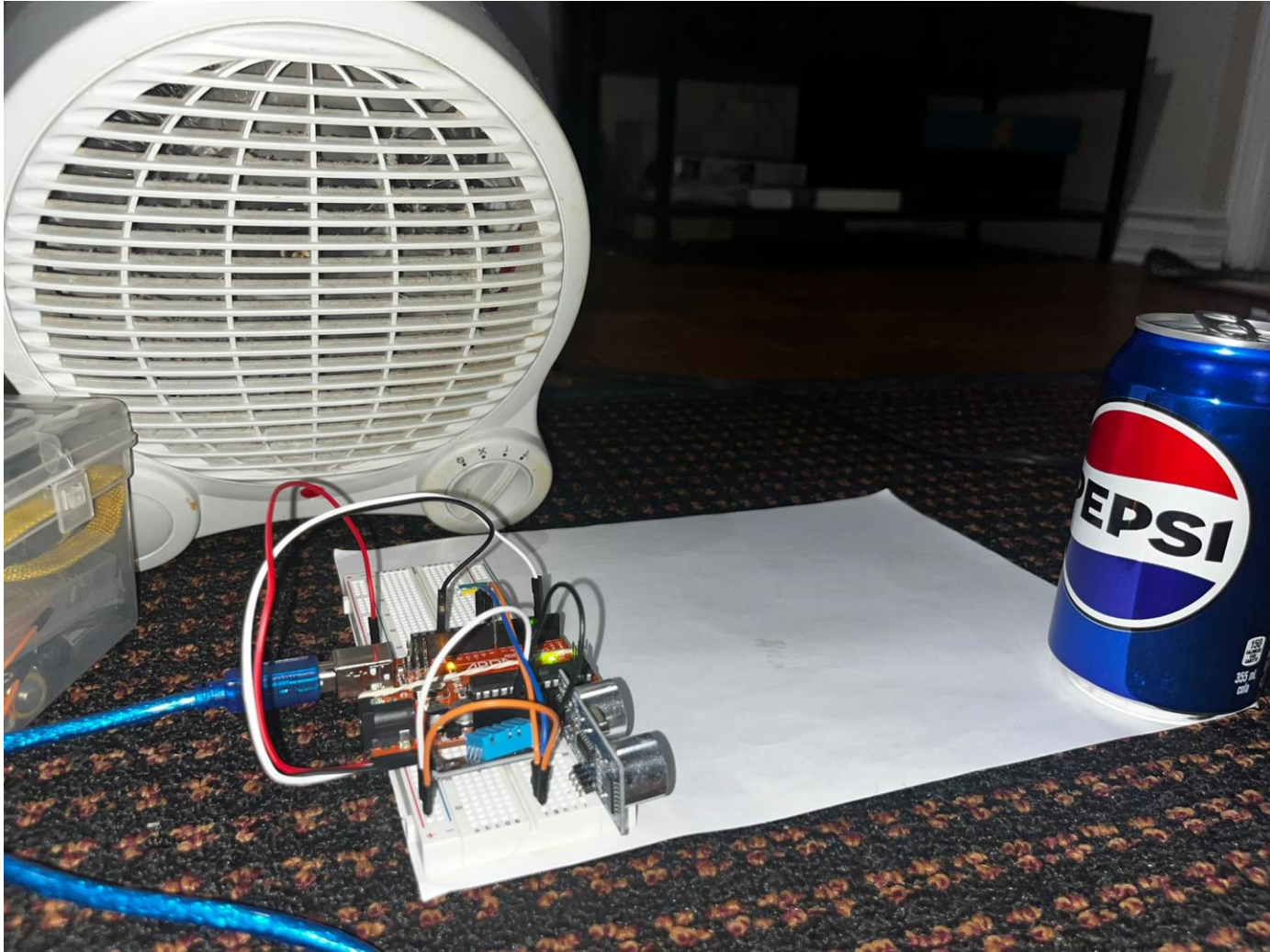


Schematic from TinkerCad

Component	Pin	Arduino Pin
DHT11	VCC	5V
	DATA	D2
	GND	GND
HC-SR04	VCC	5V
	TRIG	D8
	ECHO	D9
	GND	GND

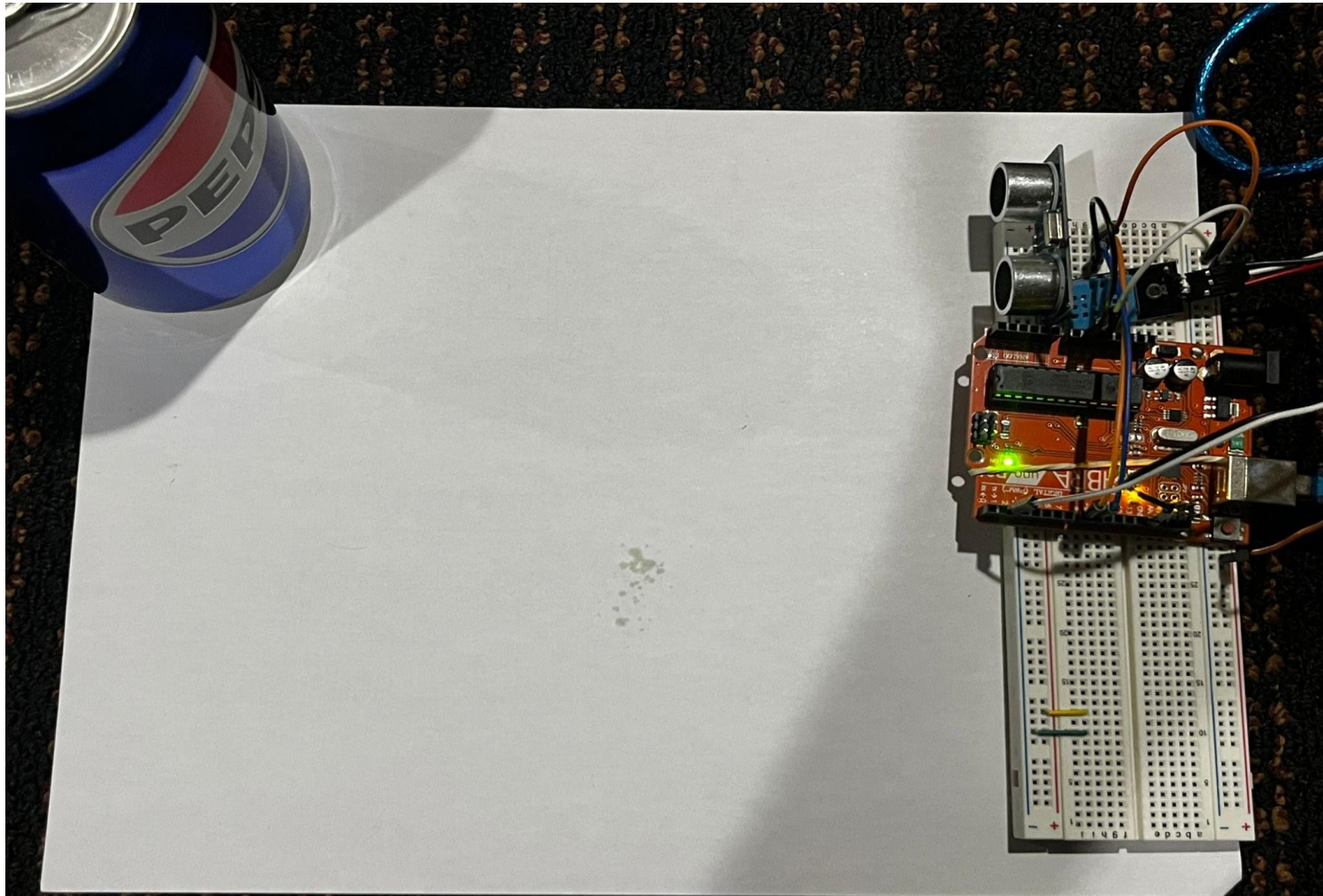
Wiring Table

Testing: High Temperature



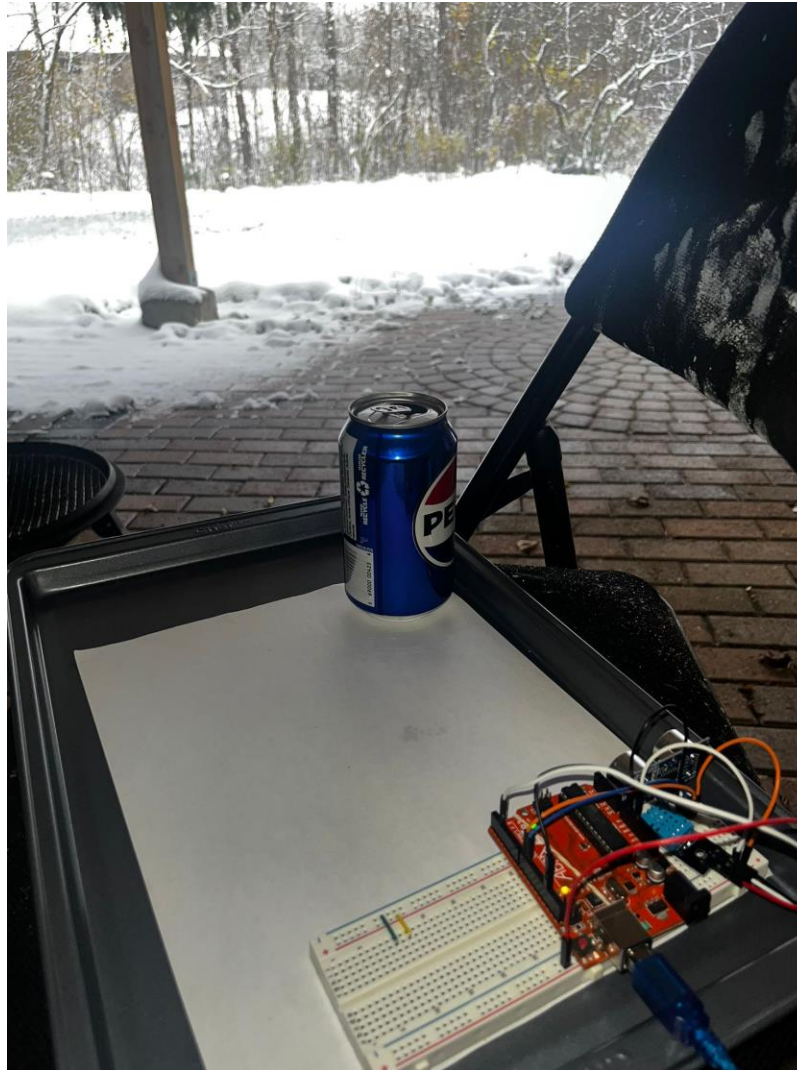
Picture of testing at room temperature

Testing: Room Temperature



Picture of testing at room temperature

Testing: Low Temperature



Picture of testing at room temperature

Testing: Results

```
16:18:22.139 -> -----  
16:18:24.122 -> Temperature: 40.60 °C | Humidity: 14.00 %  
16:18:24.186 -> Distance: 16.14 cm  
16:18:24.186 -> -----  
16:18:26.198 -> Temperature: 40.80 °C | Humidity: 14.00 %  
16:18:26.233 -> Distance: 16.16 cm  
16:18:26.263 -> -----  
16:18:28.250 -> Temperature: 40.20 °C | Humidity: 14.00 %  
16:18:28.315 -> Distance: 16.09 cm  
16:18:28.315 -> -----
```



At High
Temperature

```
01:33:48.395 -> -----  
01:33:50.386 -> Temperature: 25.40 °C | Humidity: 48.00 %  
01:33:50.418 -> Distance: 16.93 cm  
01:33:50.451 -> -----  
01:33:52.432 -> Temperature: 25.90 °C | Humidity: 48.00 %  
01:33:52.498 -> Distance: 16.93 cm  
01:33:52.498 -> -----  
01:33:54.497 -> Temperature: 25.60 °C | Humidity: 48.00 %  
01:33:54.531 -> Distance: 16.93 cm  
01:33:54.563 -> -----
```



At Room
Temperature

```
13:22:39.496 -> -----  
13:22:41.486 -> Temperature: 6.00 °C | Humidity: 58.00 %  
13:22:41.520 -> Distance: 16.98 cm  
13:22:41.552 -> -----  
13:22:43.518 -> Temperature: 6.00 °C | Humidity: 58.00 %  
13:22:43.584 -> Distance: 16.98 cm  
13:22:43.621 -> -----  
13:22:45.587 -> Temperature: 6.40 °C | Humidity: 58.00 %  
13:22:45.619 -> Distance: 16.98 cm  
13:22:45.651 -> -----
```



At Low
Temperature

Methodology

Analogy: Like diabetes tests vary by meal timing, location, and population, ultrasonic readings vary by temperature, surface, alignment, and power.

Goal: Not to model “the whole world,” but to isolate key factors and measure their effect reliably.

Controls

- Fixed 17.8 cm setup using an A4 sheet and a Pepsi can target for consistent ultrasonic reading. (Used a ruler to measure distance manually)
- Constant supply: 5v
- DHT11 placed 2–3 cm from HC-SR04 (same air mass)

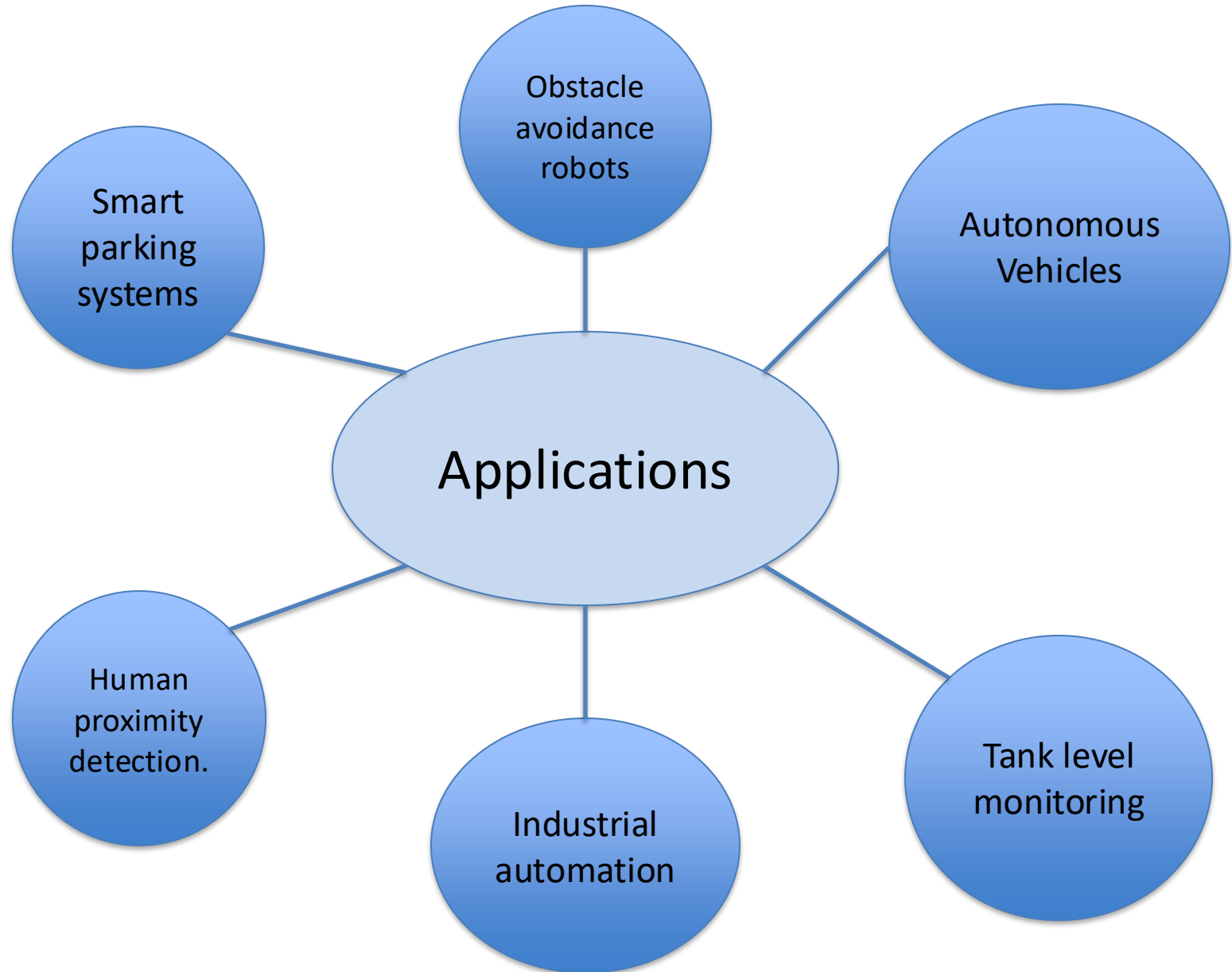
Temperature bands: ~6 °C (cool), ~25 °C (room), ~40 °C (warm)



Note: Results apply to our setup & conditions, not “all environments”.

Challenges

- During testing, it was challenging to obtain accurate low-temperature readings because ambient conditions were too warm. To achieve cold readings, the system had to be placed inside a refrigerator, requiring manual effort to hold the laptop and maintain connectivity for live monitoring.
- We had to wait for weather to be cold and then test using extended wiring.
- Power and Cable Constraints: Short cables limited placement flexibility during temperature testing.



Conclusion

The testing was successfully completed, and the HC-SR04 ultrasonic sensor performed accurately and consistently at a fixed distance of 17.8 cm

The sensor maintained consistent response time and precise signal timing.

Calibration confirmed that the sensor operates reliably within the expected range (2 – 400 cm) under controlled indoor conditions.

Temperature and environmental factors (like air movement and surface reflection) caused minor variations in readings.

Overall, the sensor demonstrated stable and reliable distance measurement, confirming its suitability for robotics and obstacle-avoidance applications.

Phase 2 Progress

- **Objective:** To integrate and test the HC-SR04 Ultrasonic Sensor and DHT11 Temperature Sensor.
- **Setup:** Used Arduino Uno for sensor interfacing and data logging.
- **Temperature Sensor Test:** Verified DHT11 readings across multiple temperature conditions (room, cold, warm).
- **Ultrasonic Sensor Prep:** Set up for distance measurement and timing calibration.
- **Software Development:** Wrote individual sensor test codes and validated serial data output.
- **Challenges:** Wiring constraints, unstable readings at startup, and limited pin availability.
- **Next Steps:** Combine both sensors for environmental calibration and performance comparison.

Phase 2 Progress

Category	Phase 1	Phase 2
Input Handling	Used <code>scanf()</code> for typed commands like <code>FWD</code> , <code>LFT</code> , <code>STOP</code> — required pressing Enter .	Uses <code>ncurses getch()</code> for real-time key detection (e.g., arrow keys / WASD) — no Enter needed .
Control Flow	Executes one command at a time and waits for next input.	Runs a continuous event loop — robot responds instantly to keypresses.
User Interface	Plain text output on terminal (no layout).	Dynamic <code>ncurses</code>-based interface showing speed, direction, and status updates.
Responsiveness	Slower; blocking input caused slight lag.	Non-blocking ; immediate motion control.
Speed Control	Two fixed speeds (Slow 70%, Fast 90%) selected by typing “1” or “2”.	Speeds adjustable dynamically with keyboard shortcuts during motion.
Direction Control	String commands like “FWD”, “BCK”, “LFT”, “RGT”.	Arrow key or WASD control for intuitive movement.
Safety Features	Manual stop function (<code>STOP</code>) before direction change.	Automatic safe-stop delay and direction handling built in.
Code Structure	Sequential <code>if-else</code> statements with static flow.	Modular input–process–actuate loop architecture with <code>ncurses</code> window refresh.
Libraries Used	<code><wiringPi.h></code> , <code><softPwm.h></code> , <code><stdio.h></code> , <code><string.h></code>	Added <code><ncurses.h></code> for terminal-based GUI and event handling.
User Experience	Command-line style, slower interaction.	Interactive, game-like control with real-time display.
Overall Improvement	Functional but basic control and interface.	Enhanced usability, responsiveness, and visual feedback

Thank You for Watching 😊

Any Questions? 🤔

Note: For additional information about the sensor, HC-SR04 Ultrasonic sensor datasheet can be found on our GitHub.

Link: [ENGG-PROJECT-3/hc-sr04 datasheet.pdf at Week-6 · Bilal2866/ENGG-PROJECT-3](https://github.com/Bilal2866/ENGG-PROJECT-3/blob/master/hc-sr04%20datasheet.pdf)